voltage amplitude V_2^+ given by

$$V_2^+ = \Gamma_g V_1^- = \Gamma_g \Gamma_L V_1^+, \tag{2.148}$$

where

$$\Gamma_g = \frac{R_g - Z_0}{R_g + Z_0} \tag{2.149}$$

is the reflection coefficient of the generator resistance R_g. For $R_g = 4Z_0$, we have $\Gamma_g = 0.6$. As time progresses after $t = 2T$, the wave V_2^+ travels down the line toward the load and adds to the previously established voltage on the line. Hence, at $t = 5T/2$, the total voltage on the first half of the line is

$$V(z, 5T/2) = V_1^+ + V_1^- + V_2^+ = (1 + \Gamma_L + \Gamma_L\Gamma_g)V_1^+ \qquad (0 \le z < l/2), \tag{2.150}$$

while on the second half of the line the voltage is only

$$V(z, 5T/2) = V_1^+ + V_1^- = (1 + \Gamma_L)V_1^+ \qquad (l/2 \le z \le l). \tag{2.151}$$

The voltage distribution is shown in **Fig. 2-43(c)**.

So far, we have examined the transient response of the voltage wave $V(z,t)$. The associated transient response of the current $I(z,t)$ is shown in **Figs. 2-43(d)–(f)**. The current behaves similarly to the voltage $V(z,t)$, except for one important difference. Whereas at either end of the line the reflected voltage is related to the incident voltage by the reflection coefficient at that end, the reflected current is related to the incident current by the negative of the reflection coefficient. This property of wave reflection is expressed by Eq. (2.61). Accordingly,

$$I_1^- = -\Gamma_L I_1^+, \tag{2.152a}$$

$$I_2^+ = -\Gamma_g I_1^- = \Gamma_g \Gamma_L I_1^+, \tag{2.152b}$$

and so on.

► The multiple-reflection process continues indefinitely, and the ultimate value that $V(z,t)$ reaches as t approaches $+\infty$ is the same at all locations on the transmission line. ◄

The final value of V is given by

$$\begin{aligned} V_\infty &= V_1^+ + V_1^- + V_2^+ + V_2^- + V_3^+ + V_3^- + \cdots \\ &= V_1^+[1 + \Gamma_L + \Gamma_L\Gamma_g + \Gamma_L^2\Gamma_g + \Gamma_L^2\Gamma_g^2 + \Gamma_L^3\Gamma_g^2 + \cdots] \\ &= V_1^+[(1 + \Gamma_L)(1 + \Gamma_L\Gamma_g + \Gamma_L^2\Gamma_g^2 + \cdots)] \\ &= V_1^+(1 + \Gamma_L)[1 + x + x^2 + \cdots], \end{aligned} \tag{2.153}$$

where $x = \Gamma_L\Gamma_g$. The series inside the square bracket is the geometric series of the function

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots \qquad \text{for } |x| < 1. \tag{2.154}$$

Hence, Eq. (2.153) can be rewritten in the compact form

$$V_\infty = V_1^+ \frac{1 + \Gamma_L}{1 - \Gamma_L\Gamma_g}. \tag{2.155}$$

Upon replacing V_1^+, Γ_L, and Γ_g with Eqs. (2.145b), (2.147), and (2.149) and simplifying the resulting expression, we obtain

$$V_\infty = \frac{V_g R_L}{R_g + R_L}. \tag{2.156}$$

The voltage V_∞ is called the ***steady-state voltage*** on the line, and its expression is exactly what we should expect on the basis of dc analysis of the circuit in **Fig. 2-43(a)**, where we treat the transmission line as simply a connecting wire between the generator circuit and the load. The corresponding ***steady-state current*** is

$$I_\infty = \frac{V_\infty}{R_L} = \frac{V_g}{R_g + R_L}. \tag{2.157}$$

2-12.2 Bounce Diagrams

Keeping track of the voltage and current waves as they bounce back and forth on the line is a rather tedious process. The ***bounce diagram*** is a graphical presentation that allows us to accomplish the same goal but with relative ease. The horizontal axes in **Figs. 2-44(a)** and **(b)** represent the position along the transmission line, while the vertical axes denote time. **Figures 2-44(a)** and **(b)** pertain to $V(z,t)$ and $I(z,t)$, respectively. The bounce diagram in **Fig. 2-44(a)** consists of a zigzag line indicating the progress of the voltage wave on the line. The incident wave V_1^+ starts at $z = t = 0$ and travels in the $+z$ direction until it reaches the load at $z = l$ at time $t = T$. At the very top of the bounce diagram, the reflection coefficients are indicated by $\Gamma = \Gamma_g$ at the generator end and by $\Gamma = \Gamma_L$ at the load end. At the end of the first straight-line segment of the zigzag line, a second line is drawn to represent the reflected voltage wave $V_1^- = \Gamma_L V_1^+$. The amplitude of each new straight-line segment equals the product of the amplitude of the preceding straight-line segment and the reflection coefficient at that end of the line. The bounce diagram for the current $I(z,t)$ in **Fig. 2-44(b)** adheres to the same principle except for the reversal of the signs of Γ_L and Γ_g at the top of the bounce diagram.

电压幅度 V_2^+ 由下式给出

$$V_2^+ = \Gamma_g V_1^- = \Gamma_g \Gamma_L V_1^+, \tag{2.148}$$

在哪里

$$\Gamma_g = \frac{R_g - Z_0}{R_g + Z_0} \tag{2.149}$$

是发电机电阻 R_g 的反射系数。对于 $R_g = 4Z_0$，我们有 $\Gamma_g = 0.6$。随着时间在 $t = 2T$ 之后进行，波 V_2^+ 沿着线路向负载传播，并添加到线路上先前建立的电压。因此，在 $t = 5T/2$ 时，线路前半部分的总电压为

$$V(z, 5T/2) = V_1^+ + V_1^- + V_2^+ = (1 + \Gamma_L + \Gamma_L\Gamma_g)V_1^+ \qquad (0 \le z < l/2), \tag{2.150}$$

而线路的后半部分电压仅为

$$V(z, 5T/2) = V_1^+ + V_1^- = (1 + \Gamma_L)V_1^+ \qquad (l/2 \le z \le l). \tag{2.151}$$

电压分布如图2-43(c)所示。

到目前为止，我们已经研究了电压波 V (z,t) 的瞬态响应。电流 I(z,t) 的相关瞬态响应如图 2 和 3 所示。 2-43(d)–(f)。除了一个重要差异外，电流的行为与电压 V(z,t) 类似。在线路的任一端，反射电压与入射电压通过该端的反射系数相关，而反射电流与入射电流通过反射系数的负值相关。波反射的这一特性可以用方程表示： (2.61)。因此，

$$I_1^- = -\Gamma_L I_1^+, \tag{2.152a}$$

$$I_2^+ = -\Gamma_g I_1^- = \Gamma_g \Gamma_L I_1^+, \tag{2.152b}$$

等等。

► The multiple-reflection process continues indefinitely, and the ultimate value that $V(z,t)$ reaches as t approaches $+\infty$ is the same at all locations on the transmission line. ◄

V 的最终值由下式给出

$$\begin{aligned} V_\infty &= V_1^+ + V_1^- + V_2^+ + V_2^- + V_3^+ + V_3^- + \cdots \\ &= V_1^+[1 + \Gamma_L + \Gamma_L\Gamma_g + \Gamma_L^2\Gamma_g + \Gamma_L^2\Gamma_g^2 + \Gamma_L^3\Gamma_g^2 + \cdots] \\ &= V_1^+[(1 + \Gamma_L)(1 + \Gamma_L\Gamma_g + \Gamma_L^2\Gamma_g^2 + \cdots)] \\ &= V_1^+(1 + \Gamma_L)[1 + x + x^2 + \cdots], \end{aligned} \tag{2.153}$$

其中 $x = \Gamma_L\Gamma_g$。方括号内的级数是函数的几何级数

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots \qquad \text{for } |x| < 1. \tag{2.154}$$

因此，等式。 (2.153) 可以重写为紧凑形式

$$V_\infty = V_1^+ \frac{1 + \Gamma_L}{1 - \Gamma_L\Gamma_g}. \tag{2.155}$$

将 V_1^+、Γ_L 和 Γ_g 替换为等式后： (2.145b)、(2.147) 和 (2.149) 并简化结果表达式，我们得到

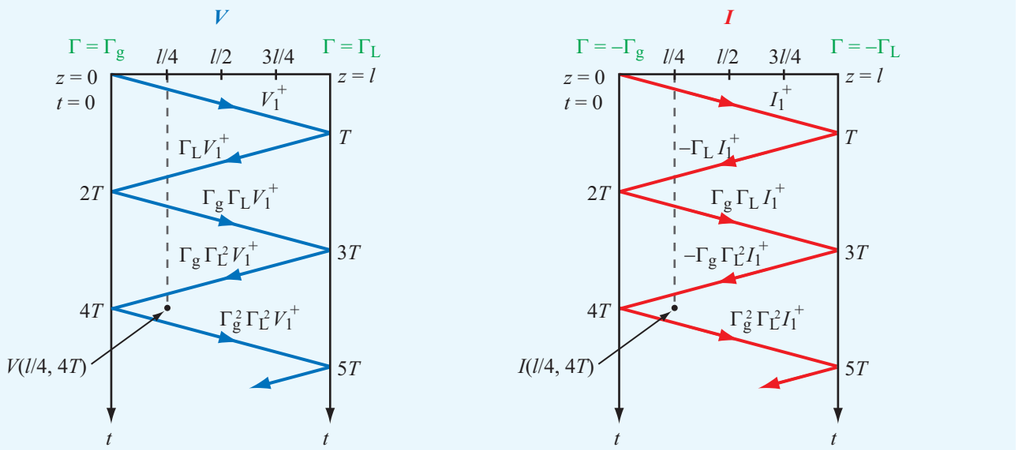
$$V_\infty = \frac{V_g R_L}{R_g + R_L}. \tag{2.156}$$

电压V_∞被称为线路上的稳态电压，其表达式正是我们根据图2-43(a)中电路的直流分析所期望的，其中我们将传输线视为发电机电路和负载之间的简单连接线。相应的稳态电流为

$$I_\infty = \frac{V_\infty}{R_L} = \frac{V_g}{R_g + R_L}. \tag{2.157}$$

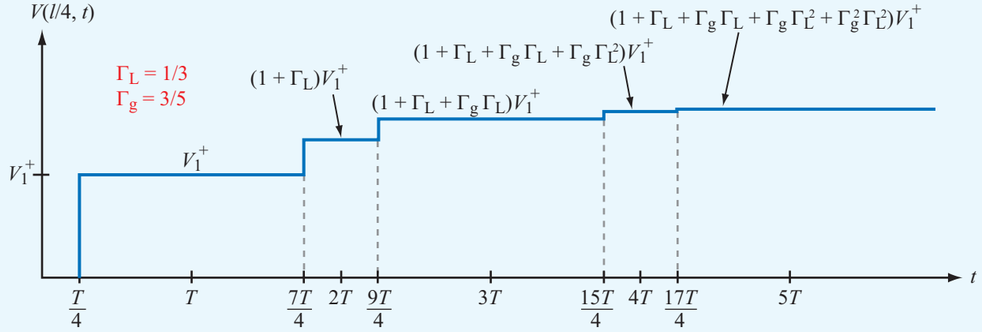
2-12.2 弹跳图

跟踪在线路上来回反弹的电压和电流波是一个相当乏味的过程。反弹图是一种图形表示，它使我们能够相对轻松地实现相同的目标。图中的水平轴。图2-44(a)和(b)表示沿着传输线的位置，而纵轴表示时间。图 2-44(a) 和 (b) 分别涉及 V (z,t) 和 I(z,t)。图2-44(a)中的反弹图由一条锯齿线组成，指示线上电压波的进展。入射波 V_1^+ 从 z = t = 0 开始，沿 +z 方向行进，直到在时间 t = T 到达 z = l 处的载荷。在弹跳图的最顶部，反射系数在发电机端用 $\Gamma = \Gamma_g$ 表示，在负载端用 $\Gamma = \Gamma_L$ 表示。在锯齿形线的第一条直线段的末端，绘制第二条线来表示反射电压波 $V_1^- = \Gamma_L V_1^+$。每个新直线段的振幅等于前一个直线段的振幅与该直线末端的反射系数的乘积。图2-44(b)中的电流I(z,t)的反弹图遵循相同的原理，除了反弹图顶部的Γ_L和Γ_g的符号相反。



(a) Voltage bounce diagram

(b) Current bounce diagram



(c) Voltage versus time at $z = l/4$

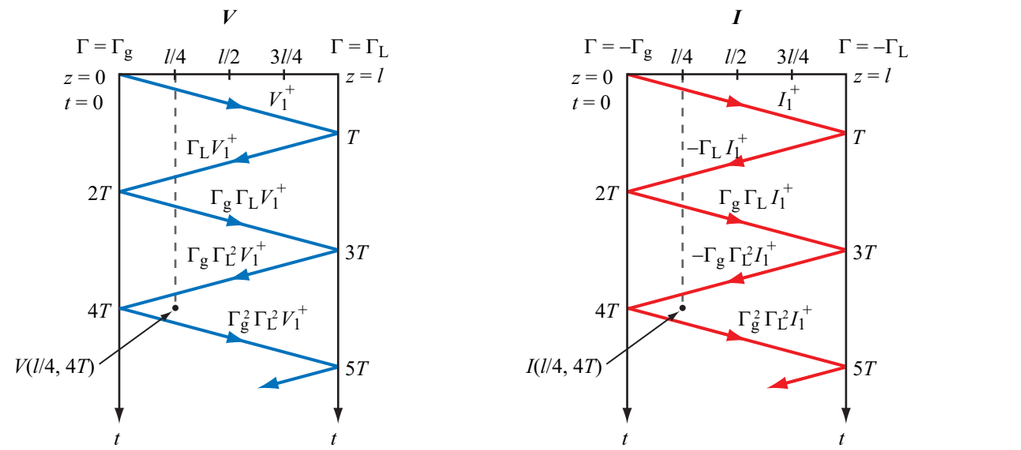
Figure 2-44 Bounce diagrams for (a) voltage and (b) current. In (c), the voltage variation with time at $z = l/4$ for a circuit with $\Gamma_g = 3/5$ and $\Gamma_L = 1/3$ is deduced from the vertical dashed line at $l/4$ in (a).

Using the bounce diagram, the total voltage (or current) at any point z_1 and time t_1 can be determined by first drawing a vertical line through point z_1, and then adding the voltages (or currents) of all the zigzag segments intersected by that line between $t = 0$ and $t = t_1$. To find the voltage at $z = l/4$ and $T = 4T$, for example, we draw a dashed vertical line in **Fig. 2-44(a)** through $z = l/4$, and we extend it from $t = 0$ to $t = 4T$. The dashed line intersects four line segments. The total voltage at $z = l/4$ and $t = 4T$ therefore is

$$V(l/4, 4T) = V_1^+ + \Gamma_L V_1^+ + \Gamma_g \Gamma_L V_1^+ + \Gamma_g \Gamma_L^2 V_1^+$$
$$= V_1^+ (1 + \Gamma_L + \Gamma_g \Gamma_L + \Gamma_g \Gamma_L^2).$$

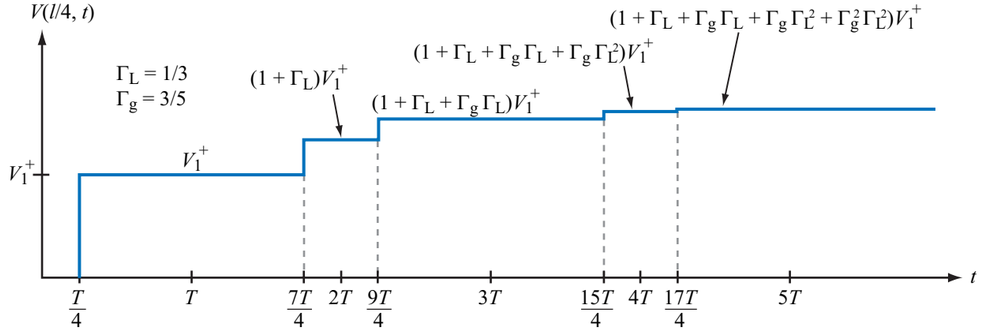
The time variation of $V(z,t)$ at a specific location z can be obtained by plotting the values of $V(z,t)$ along the (dashed)

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(a) Voltage bounce diagram

(b) Current bounce diagram



(c) Voltage versus time at $z = l/4$

图 2-44 (a) 电压和 (b) 电流的跳动图。在 (c) 中，对于具有 $\Gamma_g = 3/5$ 和 $\Gamma_L = 1/3$ 的电路，z = l/4 处的电压随时间的变化是从 (a) 中 l/4 处的垂直虚线推导出来的。

使用反弹图，可以通过首先画一条穿过点 z_1 的垂直线，然后将 t = 0 和 $t = t_1$ 之间与该线相交的所有锯齿形段的电压（或电流）相加来确定任意点 z_1 和时间 t_1 处的总电压（或电流）。例如，为了求 z = l/4 和 T = 4T 处的电压，我们在图 2-44(a) 中通过 z = l/4 画一条垂直虚线，并将其从 t = 0 延伸到

t = 4T 。虚线与四个线段相交。因此 z = l/4 和 t = 4T 处的总电压为

$$V(l/4, 4T) = V_1^+ + \Gamma_L V_1^+ + \Gamma_g \Gamma_L V_1^+ + \Gamma_g \Gamma_L^2 V_1^+$$
$$= V_1^+ (1 + \Gamma_L + \Gamma_g \Gamma_L + \Gamma_g \Gamma_L^2).$$

通过沿（虚线）绘制 V (z,t) 值，可以获得特定位置 z 处 V (z,t) 的时间变化。

vertical line passing through z. **Figure 2-44(c)** shows the variation of V as a function of time at $z = l/4$ for a circuit with $\Gamma_g = 3/5$ and $\Gamma_L = 1/3$.

Example 2-16: Pulse Propagation

The transmission-line circuit of **Fig. 2-45(a)** is excited by a rectangular pulse of duration $\tau = 1$ ns that starts at $t = 0$. Establish the waveform of the voltage response at the load given that the pulse amplitude is 5 V, the phase velocity is c, and the length of the line is 0.6 m.

Solution: The one-way propagation time is

$$T = \frac{l}{c} = \frac{0.6}{3 \times 10^8} = 2 \text{ ns}.$$

The reflection coefficients at the load and the sending end are

$$\Gamma_L = \frac{R_L - Z_0}{R_L + Z_0} = \frac{150 - 50}{150 + 50} = 0.5,$$
$$\Gamma_g = \frac{R_g - Z_0}{R_g + Z_0} = \frac{12.5 - 50}{12.5 + 50} = -0.6.$$

By Eq. (2.143), the pulse is treated as the sum of two step functions: one that starts at $t = 0$ with an amplitude $V_{10} = 5$ V and a second one that starts at $t = 1$ ns with an amplitude $V_{20} = -5$ V. Except for the time delay of 1 ns and the sign reversal of all voltage values, the two step functions generate identical bounce diagrams, as shown in **Fig. 2-45(b)**. For the first step function, the initial voltage is given by

$$V_1^+ = \frac{V_{01} Z_0}{R_g + Z_0} = \frac{5 \times 50}{12.5 + 50} = 4 \text{ V}.$$

Using the information displayed in the bounce diagram, it is straightforward to generate the voltage response shown in **Fig. 2-45(c)**. Note that at $t = 2$ ns, the voltage at the load consists of both $V_1^+ = 4$ V, and its reflection $V_1^- = \Gamma_L V_1^+ = 2$ V. Thus, the total voltage at the load at $t = 2$ ns is 6 V. A similar process occurs at 3 ns due to the negative step function, which generates -6 V. Consequently, the sum of the two step-function contributions adds up to zero and stays that way until $t = 6$ ns.

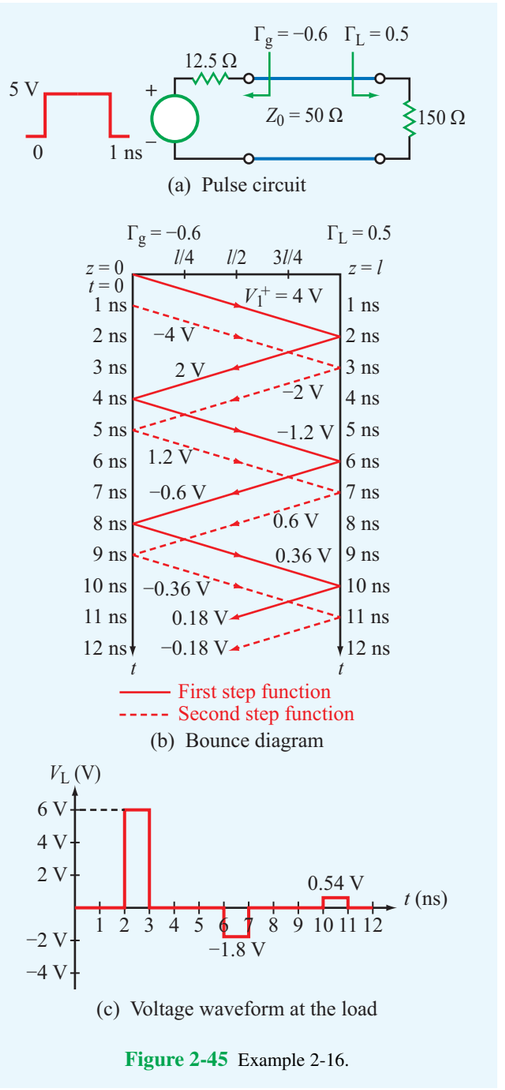


Figure 2-45 Example 2-16.

Example 2-17: Time-Domain Reflectometer

A time-domain reflectometer (TDR) is an instrument used to locate faults on a transmission line. Consider, for example, a (formerly matched) long underground or undersea cable that gets damaged at some distance d from the sending end of the line. The damage may alter the electrical properties or the shape of the cable, causing it to exhibit at the fault location an ***effective resistance*** R_{Lf}. A TDR sends a step voltage down

通过 z 的垂直线。图 2-44(c) 显示了具有 $\Gamma_g = 3/5$ 和 $\Gamma_L = 1/3$ 的电路在 z = l/4 处 V 随时间的变化。

例 2-16：脉冲传播

图 2-45(a) 的传输线电路由从 t = 0 开始、持续时间为 τ = 1 ns 的矩形脉冲激励。假设脉冲幅度为 5 V、相速度为 c 、线路长度为 0.6 m，则建立负载处的电压响应波形。

解：单向传播时间为

$$T = \frac{l}{c} = \frac{0.6}{3 \times 10^8} = 2 \text{ ns}.$$

负载端和发送端的反射系数为

$$\Gamma_L = \frac{R_L - Z_0}{R_L + Z_0} = \frac{150 - 50}{150 + 50} = 0.5,$$
$$\Gamma_g = \frac{R_g - Z_0}{R_g + Z_0} = \frac{12.5 - 50}{12.5 + 50} = -0.6.$$

由方程式在方程（2.143）中，脉冲被视为两个阶跃函数的和：一个从 t = 0 开始，幅度为 $V_{10} = 5$ V，第二个从 t = 1 ns 开始，幅度为 $V_{20} = -5$ V。除了 1 ns 的时间延迟和所有电压值的符号反转之外，这两个阶跃函数生成相同的反弹图，如图 1 所示。 2-45(b)。对于第一步函数，初始电压由下式给出

$$V_1^+ = \frac{V_{01} Z_0}{R_g + Z_0} = \frac{5 \times 50}{12.5 + 50} = 4 \text{ V}.$$

使用反弹图中显示的信息，可以直接生成图 2-45(c) 中所示的电压响应。请注意，在 t = 2 ns 时，负载上的电压由 $V_1^+ = 4$ V 及其反射 $V_1^- = \Gamma_L V_1^+ = 2$ V 组成。因此，在 t = 2 ns 时负载上的总电压为 6 V。由于负阶跃函数，在 3 ns 时会发生类似的过程，生成 −6 V。因此，两个阶跃函数贡献的总和加上直到 0 并保持这种状态直到 t = 6 ns。

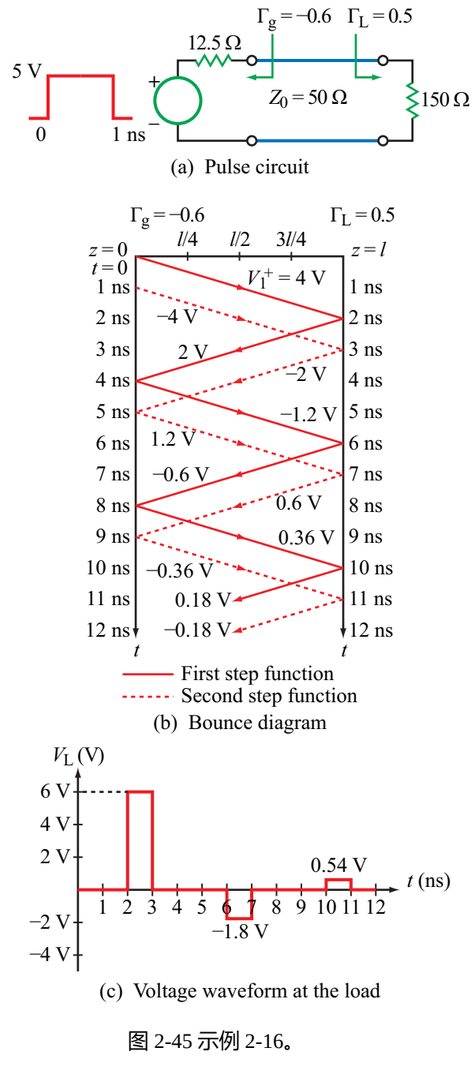
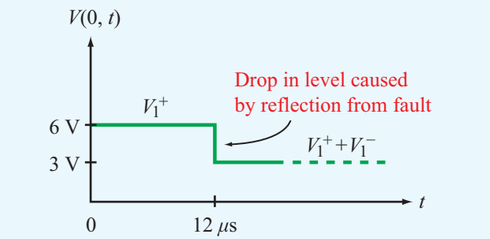


图 2-45 示例 2-16。

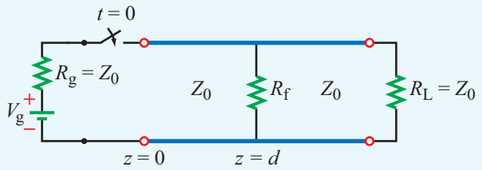
例 2-17：时域反射计

时域反射计 (TDR) 是一种用于定位传输线路故障的仪器。例如，考虑一条（以前匹配的）长地下或海底电缆

在距线路发送端一定距离 d 处损坏。损坏可能会改变电缆的电气特性或形状，导致其在故障位置表现出有效电阻 R_{Lf}。 TDR 发送阶跃电压下降



(a) Observed voltage at the sending end



(b) The fault at $z = d$ is represented by a fault resistance R_f

Figure 2-46 Time-domain reflectometer of Example 2-17.

the line, and by observing the voltage at the sending end as a function of time, it is possible to determine the location of the fault and its severity.

If the voltage waveform shown in **Fig. 2-46(a)** is seen on an oscilloscope connected to the input of a 75 Ω matched transmission line, determine (a) the generator voltage, (b) the location of the fault, and (c) the fault shunt resistance. The line's insulating material is Teflon with $\varepsilon_r = 2.1$.

Solution: **(a)** Since the line is properly matched, $R_g = R_L = Z_0$. In **Fig. 2-46(b)**, the fault located a distance d from the sending end is represented by a shunt resistance R_f. For a matched line, Eq. (2.145b) gives

$$V_1^+ = \frac{V_g Z_0}{R_g + Z_0} = \frac{V_g Z_0}{2Z_0} = \frac{V_g}{2}.$$

According to **Fig. 2-46(a)**, $V_1^+ = 6$ V. Hence,

$$V_g = 2V_1^+ = 12 \text{ V}.$$

(b) The propagation velocity on the line is

$$u_p = \frac{c}{\sqrt{\varepsilon_r}} = \frac{3 \times 10^8}{\sqrt{2.1}} = 2.07 \times 10^8 \text{ m/s}.$$

For a fault at a distance d, the round-trip time delay of the echo is

$$\Delta t = \frac{2d}{u_p}.$$

From **Fig. 2-46(a)**, $\Delta t = 12$ μs. Hence,

$$d = \frac{\Delta t}{2} u_p = \frac{12 \times 10^{-6}}{2} \times 2.07 \times 10^8 = 1{,}242 \text{ m}.$$

(c) The drop in level of $V(0,t)$ shown in **Fig. 2-46(a)** represents V_1^-. Thus,

$$V_1^- = \Gamma_f V_1^+ = -3 \text{ V},$$

or

$$\Gamma_f = \frac{-3}{6} = -0.5,$$

where Γ_f is the reflection coefficient due to the effective fault resistance R_{Lf} that appears at $z = d$.

From Eq. (2.59),

$$\Gamma_f = \frac{R_{Lf} - Z_0}{R_{Lf} + Z_0},$$

which leads to $R_{Lf} = 25$ Ω. This fault load is composed of the parallel combination of the fault shunt resistance R_f and the characteristic impedance Z_0 of the line to the right of the fault:

$$\frac{1}{R_{Lf}} = \frac{1}{R_f} + \frac{1}{Z_0},$$

so the shunt resistance must be 37.5 Ω.

Example 2-18: Pulse Transmission

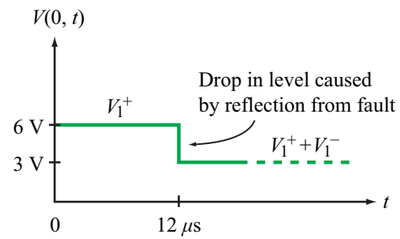
When excited by a pulse source of amplitude 1 V and duration 5 μs, a transmission line—with unknown characteristic impedance Z_0 and unknown load resistance R_L—exhibits the voltage response shown in **Fig. 2-47(b)** when observed at the middle of the line. The line relative permittivity is $\varepsilon_r = 2.25$. Determine (a) the length of the line, (b) Z_0, and (c) R_L.

Solution: **(a)** The phase velocity is

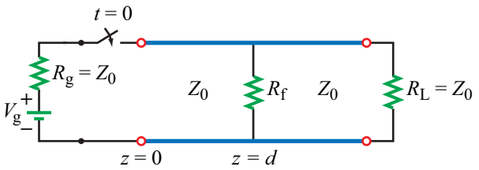
$$u_p = \frac{c}{\sqrt{\varepsilon_r}} = \frac{3 \times 10^8}{\sqrt{2.25}} = 2 \times 10^8 \text{ m/s}.$$

It takes 10 μs for the initial part of the pulse to reach the middle of the line, so it would take it $T = 20$ μs to reach the end of the line. Hence,

$$l = u_p T = 2 \times 10^8 \times 20 \times 10^{-6} = 4000 \text{ m}.$$



(a) Observed voltage at the sending end



(b) The fault at $z = d$ is represented by a fault resistance R_f

图 2-46 例 2-17 的时域反射计。

线路，并通过观察发送端的电压随时间的变化，可以确定故障的位置及其严重程度。

如果在连接到 75 Ω 匹配传输线输入的示波器上看到如图 2-46(a) 所示的电压波形，则确定 (a) 发电机电压、(b) 故障位置和 (c) 故障分流电阻。该线路的绝缘材料为 $\varepsilon_r = 2.1$ 的聚四氟乙烯。

解：(a) 由于线路正确匹配，因此 $R_g = R_L = Z_0$。图2-46(b)中，距发送端距离d处的故障用分流电阻R_f表示。对于匹配线，方程。(2.145b) 给出

$$V_1^+ = \frac{V_g Z_0}{R_g + Z_0} = \frac{V_g Z_0}{2Z_0} = \frac{V_g}{2}.$$

根据图2-46(a)，$V_1^+ = 6$ V。因此，

$$V_g = 2V_1^+ = 12 \text{ V}.$$

(b) 线上的传播速度为

$$u_p = \frac{c}{\sqrt{\varepsilon_r}} = \frac{3 \times 10^8}{\sqrt{2.1}} = 2.07 \times 10^8 \text{ m/s}.$$

对于距离 d 处的故障，回波的往返时间延迟为

$$\Delta t = \frac{2d}{u_p}.$$

由图2-46(a)可知，$\Delta t = 12$ μs。因此，

$$d = \frac{\Delta t}{2} u_p = \frac{12 \times 10^{-6}}{2} \times 2.07 \times 10^8 = 1{,}242 \text{ m}.$$

(c) 图2-46(a)所示的V(0,t)电平下降代表V_1^-。因此，

$$V_1^- = \Gamma_f V_1^+ = -3 \text{ V},$$

或者

$$\Gamma_f = \frac{-3}{6} = -0.5,$$

其中，Γ_f 是由出现在 z = d 处的有效故障电阻 R_{Lf} 产生的反射系数。

从方程。(2.59),

$$\Gamma_f = \frac{R_{Lf} - Z_0}{R_{Lf} + Z_0},$$

这导致 $R_{Lf} = 25$ Ω。该故障负载由故障分流电阻R_f和故障右侧线路的特性阻抗Z_0并联组合而成：

$$\frac{1}{R_{Lf}} = \frac{1}{R_f} + \frac{1}{Z_0},$$

因此分流电阻必须为 37.5 Ω。

例 2-18：脉冲传输

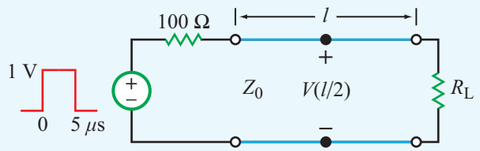
当由幅度为 1 V、持续时间为 5 μs 的脉冲源激励时，特性阻抗 Z_0 和负载电阻 R_L 未知的传输线在传输线中部观察时会表现出如图 2-47(b) 所示的电压响应。线相对介电常数为$\varepsilon_r = 2.25$。确定 (a) 线的长度、(b) Z_0 和 (c) R_L。

解：(a) 相速度为

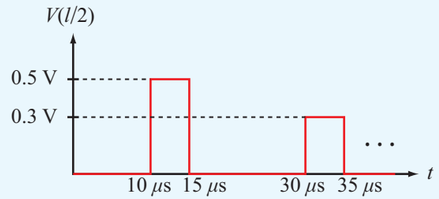
$$u_p = \frac{c}{\sqrt{\varepsilon_r}} = \frac{3 \times 10^8}{\sqrt{2.25}} = 2 \times 10^8 \text{ m/s}.$$

脉冲的初始部分需要 10 μs 才能到达线的中间，因此需要 T = 20 μs 才能到达线的末尾。因此，

$$l = u_p T = 2 \times 10^8 \times 20 \times 10^{-6} = 4000 \text{ m}.$$



(a) Transmission line



(b) Observed voltage at midpoint of the line

Figure 2-47 Circuit and voltage response of Example 2-18.

(b) From Eq. (2.145b),

$$V_1^+ = \frac{V_g Z_0}{R_g + Z_0}.$$

In the present case, $V_g = 1$ V, $R_g = 100\ \Omega$, and the waveform in **Fig. 2-47(b)** indicates that $V_1^+ = 0.5$ V. Solving for Z_0 leads to

$$Z_0 = 100\ \Omega.$$

(c) The pulse that appears between 30 μs and 35 μs is due to reflection by the load and its amplitude $V_1^- = 0.3$ V. Hence,

$$\Gamma_L = \frac{V_1^-}{V_1^+} = \frac{0.3}{0.5} = 0.6.$$

Also,

$$\Gamma_L = \frac{R_L - Z_0}{R_L + Z_0}.$$

Using $\Gamma_L = 0.6$ and $Z_0 = 100\ \Omega$ leads to

$$R_L = 400\ \Omega.$$

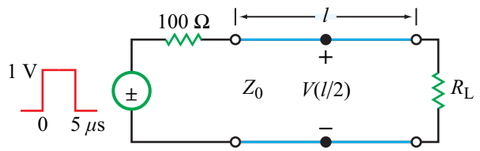
Concept Question 2-27: What is transient analysis used for?

Concept Question 2-28: The transient analysis presented in this section was for a step voltage. How does one use it for analyzing the response to a pulse?

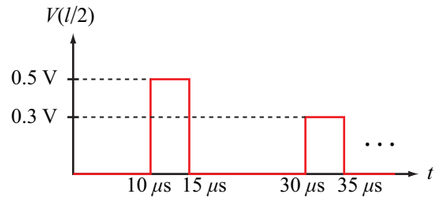
Concept Question 2-29: What is the difference between the bounce diagram for voltage and the bounce diagram for current?

Concept Question 2-30: What is a TDR and what is it used for?

Concept Question 2-31: What do V_∞ and I_∞ represent?



(a) Transmission line



(b) Observed voltage at midpoint of the line

图 2-47 例 2-18 的电路和电压响应。

(b) 从方程式(2.145b),

$$V_1^+ = \frac{V_g Z_0}{R_g + Z_0}.$$

在本例中，$V_g = 1$ V，$R_g = 100\ \Omega$，图2-47(b)中的波形表明$V_1^+ = 0.5$ V。求解Z_0导致

$$Z_0 = 100\ \Omega.$$

(c) 在 30 μs 和 35 μs 之间出现的脉冲是由于负载的反射而产生的，其幅度 $V_1^- = 0.3$ V。因此，

$$\Gamma_L = \frac{V_1^-}{V_1^+} = \frac{0.3}{0.5} = 0.6.$$

还，

$$\Gamma_L = \frac{R_L - Z_0}{R_L + Z_0}.$$

使用 $\Gamma_L = 0.6$ 和 $Z_0 = 100\ \Omega$ 会导致

$$R_L = 400\ \Omega.$$

Concept Question 2-27: What is transient analysis used for?

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